

Core java

Arrays

Theory Notes

1. Arrays

- **Definition:** A data structure that stores elements in contiguous memory locations.
- **Types:**
 - **1D Array** → linear collection of elements.
 - **2D Array** → matrix-like structure (rows × columns).
 - **Jagged Array** → array of arrays with different column sizes.
- **Access:** arr[index] (index starts at 0).
- **Iteration:**
 - for loop → index-based.
 - for-each loop → directly iterates over elements.

2. Scanner Class

- Used for **input handling** in Java.
- Common methods:
 - nextInt() → reads integer.
 - nextFloat() → reads float.
 - nextDouble() → reads double.
 - next() → reads single word.
 - nextLine() → reads entire line.
 - next().charAt(index) → reads character at position.

Coding Questions

1. Write a program to create an array of size n (user input) and read elements using Scanner. Print them using both for and for-each loops.
2. Create a 2D array (matrix) and print it in row-column format.
3. Demonstrate a jagged array where each row has different column sizes.

4. Write a program to reverse a string entered by the user.
5. Create a program to find the maximum element in an integer array.
6. Write a program to calculate the sum of all elements in an array.

```
import java.util.Scanner;

public class ArrayString {
    public static void main(String[] args) {
        // System.out.println("Arrays");

        // array -> data structure , contiguous memory

        // primitive -> stack
        // non-primitive -> heap , no-fixed size

        // let arr =[1,2,4] // javascript

        // int arr[] = {1,2,3,4,5,6}; // 1D

        // int size = arr.length;

        // System.out.println(size);

        // System.out.println(arr[0]);

        // for(int i=0; i<arr.length; i++){
        //     System.out.println(arr[i]);
        // }

        // for(int i : arr){
        //     System.out.println(i);
        // }

        // multi-demensional array // 2D

        // int multi[][] = {
        //     {1,2,3},
        //     {4,5,6},
        //     {7,8,9}
        // };

        // row = 4, col = 2

        // for(int i=0; i<3; i++){
        //     for(int j=0; j<3; j++){
        //         System.out.print(multi[i][j]+" ");
        //     }
        // }
```

```

//     }
//     System.out.println();
// }

// int multi2[][]={
//     {1,2},
//     {3,4},
//     {5,6},
//     {7,8}
// };

// for(int i=0; i<4; i++){
//     for(int j=0; j<2; j++){
//         System.out.print(multi2[i][j]+" ");
//     }
//     System.out.println();
// }

// jagged array each row has different columns

// int jagged[][] = {
//     {1,2,3},
//     {4,5},
//     {6}
// };

// System.out.println(jagged[1].length);

// for(int i=0; i<3; i++){
//     for(int j=0; j<jagged[i].length; j++){
//         System.out.print(jagged[i][j]+" ");
//     }
//     System.out.println();
// }

Scanner inp = new Scanner(System.in);

// int size = inp.nextInt();

// System.out.println(size);

// float value = inp.nextFloat();

// System.out.println(value);

```

```
// double mark = inp.nextDouble();

// System.out.println(mark);

// String name = inp.next();

// String name = inp.nextLine();
// System.out.println(name);

// char grade = inp.next().charAt(1);

// System.out.println(grade);

System.out.print("Enter array size: ");
int size = inp.nextInt();

int arr[] = new int[size];

System.out.println(arr.length);

for(int i=0; i<arr.length; i++){
    arr[i] = inp.nextInt();
}

for(int i : arr){
    System.out.println(i);
}

}
```